

AI Agent Research Compendium — Volume 3: Agent Tooling, Planning & Architecture

The most practical papers on how to build, plan, and tool-augment AI agents for production systems.

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1. Agent Architecture & Design Patterns

Agentic Design Patterns: A System-Theoretic Framework (arXiv: 2601.19752)

Five functional subsystems + 12 reusable design patterns for robust agent architectures.

Must-read for systematic architecture design.

Agentic AI: Architectures, Taxonomies, and Evaluation (arXiv: 2601.12560)

Unified taxonomy: Perception, Brain, Planning, Action, Tool Use, Collaboration. Covers MCP and evaluation. **The reference paper.**

Engineering AI Agents for Clinical Workflows (arXiv: 2602.00751)

Production reference: Clean Architecture, event-driven design, per-agent MLOps, HITL governance. **Production blueprint.**

POLARIS: Typed Planning and Governed Execution (arXiv: 2601.11816)

DAG-based planning, rubric-guided selection, validator-gated execution, compiled policy guardrails.

Toward Efficient Agents: Memory, Tool Learning, and Planning (arXiv: 2601.14192)

Surveys efficiency under fixed cost budgets, analyzing Pareto frontiers between effectiveness and cost.

Unix Philosophy Informs Agentic AI Design (arXiv: 2601.11672)

File-like abstractions and code-based specifications for composable, auditable agent interfaces.

CaveAgent: Stateful Runtime Operators (arXiv: 2601.01569)

Dual-stream architecture: persistent Python runtime as central state locus with skill injection.

InfiAgent: Infinite-Horizon Agent Framework (arXiv: 2601.03204)

Keeps reasoning context bounded regardless of task duration via file-centric state externalization.

Architecting AgentOps Needs CHANGE (arXiv: 2601.06456)

Six capabilities (Contextualize, Harmonize, Anticipate, Negotiate, Generate, Evolve) for AgentOps lifecycle management.

2. Planning & Reasoning

Why Reasoning Fails to Plan (arXiv: 2601.22311)

Step-wise reasoning struggles with long-horizon planning; future-aware lookahead with reward estimation helps. **Critical for understanding agent limitations.**

ProAct: Agentic Lookahead in Interactive Environments (arXiv: 2602.05327)

Distilling environment search into causal reasoning chains for better look-ahead agent behavior.

The Path Ahead for Agentic AI (arXiv: 2601.02749)

Surveys planning, memory, tool use, iterative reasoning with critical assessment of safety and reliability.

Architecting Agentic Communities using Design Patterns (arXiv: 2601.03624)

Enterprise distributed systems patterns applied to agent coordination, governance, and collaboration.

3. Tool Use & Function Calling

Enhancing MCP with Context-Aware Server Collaboration (arXiv: 2601.11595)

Shared Context Store enables MCP servers to coordinate autonomously. **Essential for MCP production.**

DALIA: Declarative Agentic Layer for MCP (arXiv: 2601.17435)

Declarative architectural layer with formalized capabilities, discovery protocol, and deterministic task graph construction.

ToolACE-MCP: History-Aware Routing for MCP Tools (arXiv: 2601.08276)

History-aware routers using dependency graphs and multi-turn trajectory synthesis for large-scale MCP ecosystems.

Optimizing Agentic Workflows using Meta-tools (arXiv: 2601.22037)

Bundles recurring tool call sequences into deterministic meta-tools, skipping unnecessary intermediate LLM reasoning. **Direct performance optimization.**

Beyond Static Tools: Test-Time Tool Evolution (arXiv: 2601.07641)

Agents synthesize, verify, and evolve executable tools during inference instead of relying on static libraries.

Think-Augmented Function Calling (arXiv: 2601.18282)

Embeds explicit reasoning at function and parameter levels with dynamic complexity scoring for tool calls.

4. Context Engineering

Structured Context Engineering for File-Native Agents (arXiv: 2602.05447)

9,649 experiments testing YAML, JSON, Markdown context formats. **Data-driven guidance.**

Meta Context Engineering via Agentic Skill Evolution (arXiv: 2601.21557)

Bi-level framework: meta-agent evolves context skills via crossover, base agent executes them.

CEDAR: Context Engineering for Agentic Data Science (arXiv: 2601.06606)

Structured DS prompting, separate plan and code agents, smart history rendering.

SemanticALLI: Caching Reasoning, Not Responses (arXiv: 2601.16286)

Intermediate reasoning as first-class cacheable artifacts. **Direct production cost savings.**

5. Agent Training & Self-Improvement

AutoRefine: From Trajectories to Reusable Expertise (arXiv: 2601.22758)

Dual-form expertise: specialized subagents for procedural tasks, skill patterns for static knowledge.

EvoFSM: Self-Evolution with Finite State Machines (arXiv: 2601.09465)

Evolves explicit FSM instead of free-form code rewriting — constraining flow optimization to structure.

AgentDevel: Self-Evolving Agents as Release Engineering (arXiv: 2601.04620)

Implementation-blind quality signals, flip-centered regression gating for reliable self-improvement.

6. Evaluation & Observability

Architecture-Aware Evaluation Metrics (arXiv: 2601.19583)

Links agent components (planners, memory, tool routers) to observable behaviors and diagnostic metrics.

When Agents Fail: Study of Bugs in LLM Agents (arXiv: 2601.15232)

1,187 bug reports across seven frameworks categorized by type, root cause, and effect.

MAESTRO: Multi-Agent Evaluation Suite (arXiv: 2601.00481)

Framework-agnostic execution traces for systematic reliability assessment across architectures.

40+ papers. Selected for production architecture patterns, MCP innovations, context engineering for cost optimization, and self-evolution mechanisms.